ECE555 Design Project

This is a group assignment for **three-student** groups. One submission per group is needed as a single PDF file, making sure all pages are readable. Please submit each milestone to the relevant submission area on Canvas.

Milestones (Each milestone (MS) is 25 points):

1. Complete the gate-level design of your neuron.
    a. We suggest following a modular approach. Implement the sub-modules of your neuron as separate cells (e.g. multiplier, adder and the multiplexer).
    b. Submit your schematic (for modular designs submit schematics of the sub-modules and the neuron).

2. Implement the layout of the neuron proposed in MS-1.
    a. Demonstrate that your design has correct functionality. To do this, write a testbench netlist for your neuron's netlist using VDD=0.9 V. Clearly state the inputs, the weights, and the expected outputs. Make sure to include the worst-case input combination in your tests.
    b. Submit DRC and LVS screenshots of your submodules and the neuron.
    c. Submit calibre extraction screenshots of your neuron design
    d. **Note!** Make a backup of the extracted netlist.

3. Optimize the layout of the neuron.
    a. Comment on whether you could have done a better layout or routing (e.g. possibly using only M1, M2 layers and no M3 or higher, shorter routes etc.). Identify and optimize the critical path.
    b. Optimize your design
    c. Submit DRC and LVS screenshots of your submodules and the neuron
    d. Submit calibre extraction screenshots of your neuron design
    e. Did the parasitic capacitances decrease?
    f. Submit the area of your final design (the area of the boundary).

4. Prepare a report and demonstration for your neuron design.
    a. Write a testbench netlist for your neuron's extracted netlist. Use a VDD of 0.9V. Clearly state the inputs, the weights, and the expected outputs. Make sure to include the worst-case input combination in your tests.
    b. For the worst-case input combination, show that you see the expected output.
    c. For the worst-case input combination, measure the delay between the input and the output transition. This is the **T** of your gate.
    d. Submit the **T** and **f=1/T** of your gate.